DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Improving TA-LLM dialogue flow via automatic paired trajectory construction

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SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must control conversation flow: ask follow-ups, call tools, or reject invalid requests.
SFT with expert trajectories alone has fundamental limitations in learning these skills.

1 Failing to Ask Follow-up Questions

Model skips slot-filling when required arguments are missing, leading to incomplete tool calls.

2 Hallucinated Tool Invocations

Model invents argument values instead of asking the user, producing incorrect results.

3 Failing to Reject Out-of-Scope Requests

Model attempts tool calls for functionality that does not exist, causing silent failures.

Solution → DiaTool-DPO: Use Direct Preference Optimization to learn correct dialogue flows by contrasting chosen vs. rejected state trajectories — no human labeling required.

Modeling TA-LLM Dialogue as a 5-State MDP

↻ Repeat N×

S1 — Initial
No history. Out-of-scope requests → rejection.

→

S2 — Tool Selected w/o Complete Slots
Tool identified but params missing. Slot-filling QA here.

→

S3 — Tool Selected w/ Complete Slots
Tool + all argument values determined.

→

S4 — Wait for Tool Response
Tool call executed, awaiting results.

→

S5 — Complete
Tool results received, completion generated.

State Trajectories by Query Type:

Type 1 (All info provided):	S1 → S3 → S4 → S5
Type 2 (Missing arguments):	S1 → (S2 × N) → S3 → S4 → S5
Type 3 (Out-of-scope):	S1 → S1 (reject)

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1 (Sufficient Info)	1 → 3 → 4 → 5	All required arguments present. No slot-filling needed.	"Book a flight from NYC to LA on Dec 25 for 2 adults"
Type 2 (Incomplete Info)	1 → (2×N) → 3 → 4 → 5	Missing arguments. Slot-filling QA repeats N times.	"Book a flight to LA" → asks: date? passengers?
Type 3 (Out-of-Scope)	1 → 1	No suitable tool available. Must reject the tool call.	"What's the meaning of life?" → politely rejects

Why Query Types Matter for DPO

- Each query type defines a unique correct trajectory — deviations become rejected trajectories for DPO training.
- Type 1 errors: Redundant slot-filling (asking for information already provided).
- Type 2 errors: Slot hallucination (inventing argument values instead of asking the user).
- Type 3 errors: Hallucinated tool calls (attempting to use nonexistent tools).
- The MDP formulation enables systematic enumeration of all possible error trajectories per type.

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Diff.
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	E/H
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	E/H
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

Key Design Principle

Each pair teaches a specific lesson about dialogue control. Chosen trajectories follow the correct MDP path; rejected trajectories represent systematic deviations. No human annotation required — pairs are generated automatically from the MDP state graph.

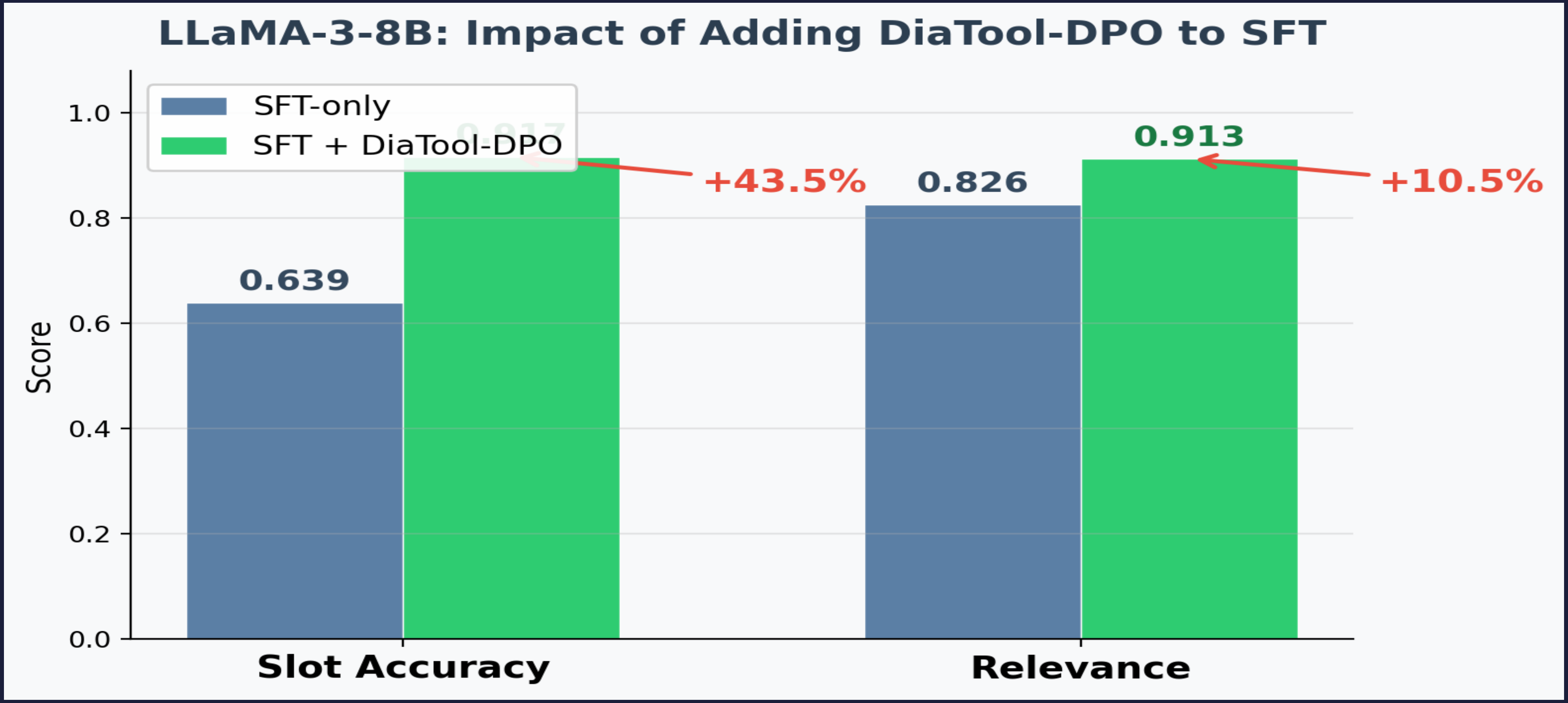
SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

FunctionChat-Bench evaluation across four metrics | Bold = best per model group

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833 ★	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900 ★	0.916	0.694	0.913 ★	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929 ★	0.833 ★	0.826	0.884 ★	0.868 ★
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771 ★	0.817	0.750	0.826 ★	0.790 ★	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871 ★	0.833 ★	0.826 ★	0.765	0.818 ★
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957 ★	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857 ★	0.929	0.917 ★	0.913 ★	0.905 ★	0.904 ★

★ = Best in group | SFT + DiaTool-DPO achieves the highest Macro Avg. for every model, confirming DPO consistently improves balanced performance across all metrics.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance



Slot: 0.639 → 0.917 (+43.5%)
Relevance: 0.826 → 0.913 (+10.5%)

Near GPT-4o performance: 94.8% information gathering, 91% tool call rejection

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1. **MDP Formulation Enables Systematic Pair Construction**

   Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation — making the approach scalable and repeatable.

2. **DPO Complements SFT — Not Replaces It**

   DiaTool-DPO-only performs poorly across all models. The method works as a post-SFT alignment step that teaches nuanced dialogue control SFT cannot capture alone.

3. **Slot-Filling and Tool Rejection See the Largest Gains**

   LLaMA-3-8B: +43.5% Slot accuracy, +10.5% Relevance. These are exactly the failure modes that SFT struggles with — confirming the DPO approach targets the right weaknesses.

4. **Language-Agnostic and Production-Ready**

   Primary experiments in Korean, validated in English. Opens path to GPT-4o-level conversational tool use on smaller, deployable open-source models.

5. **Automatic Dataset Construction Eliminates Human Labeling**

   Preference pairs are derived from the MDP state graph, not human judgment. This removes the primary bottleneck in RLHF-style alignment for tool-use scenarios.